The Complete Treatise on the General N-Institutions Model

A Typology-First Approach

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Abstract

This treatise proposes the General N-Institutions Model, a framework for the systematic classification, causal analysis, welfare evaluation, and valuation of heterogeneous institutions. Its central methodological claim is that institutional analysis should begin with the effects of institutional arrangements on the welfare of individuals inside and outside their boundaries, rather than presuming that an institution's name, legal form, stated purpose, or organizational category determines its social value.

The framework integrates institutional economics, behavioural economics, causal inference, welfare economics, econometrics, financial valuation, decision theory, law, political science, psychology, statistics, computer science, artificial intelligence, cybersecurity, and interdisciplinary risk analysis. It distinguishes observable behaviour from latent welfare, association from causation, institutional purpose from institutional consequences, and financial value from social value.

For an institution I, the framework defines individual-level potential outcomes, behavioural records, counterfactual welfare effects, inside–outside classifications, distributional measures, and institution-specific valuation vectors. It generalizes these objects to a system of N interacting institutions represented by a directed network. It further specifies identification assumptions, econometric estimators, sensitivity analyses, ethical constraints, and an empirical validation programme.

Several mathematical propositions establish properties of the classification system under explicit assumptions. These results are distinguished from empirical hypotheses and normative choices. The framework does not assume that welfare is perfectly observable, that all individuals share a common utility scale, or that heterogeneous institutions can always be ranked by one scalar.

The principal contribution is a common analytical procedure for heterogeneous institutions without imposing substantive equivalence upon them.

The treatise ends with "The End"

Keywords: institutional economics, welfare economics, behavioural economics, causal inference, institutional valuation, econometrics, social choice, financial planning and analysis, artificial intelligence, institutional networks, governance, cybersecurity, public policy.

Contents

1 Introduction **4**
- 1.1 The General Problem . . . 4
- 1.2 The Central Ordering Principle . . . 5
- 1.3 The Typology-First Requirement . . . 5
- 1.4 Scope and Epistemic Status . . . 6

2 Definitions and Basic Notation **6**
- 2.1 Institutions and Institutional Systems . . . 6
- 2.2 Individuals and Exposure . . . 6
- 2.3 Time, State, and Counterfactual . . . 7
- 2.4 The Institutional Unit of Analysis . . . 7

3 Behaviour as an Empirical Entry Point **7**
3.1 Behavioural Records . . . 7
3.2 Behavioural Policies . . . 8
3.3 Why Observed Behaviour Does Not Identify Welfare . . . 8
3.4 Behaviour, Mechanism, and Welfare . . . 9

4 The Welfare-First Classification Principle **9**
4.1 Potential Outcomes . . . 9
4.2 Three-Way Classification . . . 9
4.3 Inside and Outside Populations . . . 10
4.4 The Nine-Cell Classification . . . 10
4.5 Proposition 1: Exhaustiveness of the Three-Way Rule . . . 11
4.6 Proposition 2: Non-Identification of Individual Benefit from Aggregate Benefit . . . 11
4.7 Welfare-First Institutional Classification Principle . . . 11

5 Defining and Measuring Welfare **11**
5.1 Welfare Is a Construct, Not a Raw Observation . . . 11
5.2 A Multidimensional Welfare Representation . . . 12
5.3 Subjective and Objective Indicators . . . 12
5.4 Non-Compensable Constraints . . . 13

6 Causal Inference and Counterfactual Identification **13**
6.1 Potential Outcomes . . . 13
6.2 Target Estimands . . . 13
6.3 Identification Assumptions . . . 13
6.4 The G-Computation Formula . . . 14
6.5 Difference-in-Differences . . . 14
6.6 Selection, Exit, and Survivorship . . . 15
6.7 Spillovers and Interference . . . 15

7 Econometric Measurement and Statistical Uncertainty **15**
7.1 A General Panel Specification . . . 15
7.2 Measurement Models . . . 16
7.3 Uncertainty-Aware Classification . . . 16
7.4 Robustness and Sensitivity Analysis . . . 16

8 Behavioural Economics, Psychology, and Learning **17**
8.1 Bounded Rationality . . . 17
8.2 Institutional Feedback . . . 17
8.3 Learning and Adaptation . . . 18
8.4 Psychiatric and Psychological Evidence . . . 18

9 Institutional Purpose, Legitimacy, and Law **18**
9.1 Purpose Is Not Identical to Effect . . . 18
9.2 Legitimacy as a Distinct Dimension . . . 19
9.3 Rights and Human-Rights Due Diligence . . . 19
9.4 Legal Classification . . . 19

10 Social Choice and Aggregation **19**
10.1 The Aggregation Problem . . . 19
10.2 Distributional Welfare . . . 20
10.3 Compensation and Pareto Comparisons . . . 20
10.4 Arrow's Impossibility Result . . . 20
10.5 A Distributional Reporting Vector . . . 20

11 The General N-Institutions System **21**
11.1 Institutional Networks . . . 21
11.2 Graphical Representation . . . 21
11.3 Cross-Institutional Exposure . . . 21
11.4 Interaction Effects . . . 22
11.5 Institutional Complementarity and Substitution . . . 22

12 A Formal General N-Institutions Model **22**
12.1 Definition . . . 22
12.2 A Hierarchy of Questions . . . 23
12.3 Proposition 3: Conditional Comparability . . . 23
12.4 Institutional Non-Equivalence Principle . . . 23

13 Institutional Valuation **23**
13.1 Valuation Is Not a Single Universal Quantity . . . 23
13.2 Financial Valuation . . . 24
13.3 Stochastic Discount Factors . . . 25
13.4 Public Institutions and Social Cost-Benefit Analysis . . . 25
13.5 Financial Planning and Analysis . . . 25
13.6 Risk-Adjusted and Scenario-Based Valuation . . . 25

14 Physics, Chemistry, Biology, and Complex Systems **26**
14.1 Physics: State, Dynamics, and Stability . . . 26
14.2 Feedback and Stability . . . 26
14.3 Chemistry: Stocks, Flows, and Conservation . . . 26
14.4 Biology: Adaptation and Ecological Interdependence . . . 26
14.5 Environmental Externalities . . . 27

15 Political Science, Geopolitics, and Diplomacy **27**
15.1 Political Institutions . . . 27
15.2 Geopolitical Spillovers . . . 27
15.3 Diplomacy and Institutional Cooperation . . . 27
15.4 Warfare Economics and Security . . . 28

16 Computer Science, Data Science, and Algorithms **28**
16.1 The Institutional Data Pipeline . . . 28
16.2 Pseudocode for Institutional Classification . . . 29
16.3 Data Quality and Missingness . . . 29
16.4 Reproducible Research . . . 30
16.5 Model Validation . . . 30

17 Artificial Intelligence and Machine Learning **30**
17.1 Behavioural Prediction . . . 30
17.2 Reinforcement Learning . . . 31
17.3 AI-Assisted Institutional Evaluation . . . 31
17.4 Model Risk and Distribution Shift . . . 31

18 Cybersecurity, Privacy, and Adversarial Behaviour **32**
18.1 Institutional Data as a Security Asset . . . 32
18.2 Adversarial Data and Strategic Manipulation . . . 32
18.3 Threat Modelling . . . 32

19 An Interdisciplinary Integration Architecture **33**
19.1 Fields and Their Analytical Roles . . . 33
19.2 The Integration Rule . . . 33

20 A General Institutional Evaluation Protocol **34**
20.1 Stage One: Define the Institution . . . 34
20.2 Stage Two: Map the Affected Population . . . 34
20.3 Stage Three: Measure Behaviour and Outcomes . . . 34
20.4 Stage Four: Specify the Counterfactual . . . 34
20.5 Stage Five: Identify Causal Effects . . . 35
20.6 Stage Six: Classify Welfare Effects . . . 35
20.7 Stage Seven: Evaluate Purpose, Law, and Legitimacy . . . 35
20.8 Stage Eight: Perform Institution-Specific Valuation . . . 35
20.9 Stage Nine: Report Uncertainty and Limitations . . . 35

21 Empirical Research Programme **35**
21.1 Research Questions . . . 35
21.2 Proposed Validation Design . . . 36
21.3 Baseline Comparisons . . . 36
21.4 Falsification and Failure Conditions . . . 37

22 Limitations and Open Problems **37**
22.1 The Measurement Problem . . . 37
22.2 The Identification Problem . . . 37
22.3 The Boundary Problem . . . 37
22.4 The Aggregation Problem . . . 37
22.5 The Reflexivity Problem . . . 37
22.6 The Computational Problem . . . 37

23 Conclusions **38**

A Formal Definitions and Additional Results **38**
A.1 Definition: Institutional Welfare Profile . . . 38
A.2 Proposition 4: A Positive Average Does Not Establish a Positive Median . . . 39
A.3 Proposition 5: Classification Depends on the Counterfactual . . . 39
A.4 Proposition 6: Prediction Does Not Establish Causation . . . 39
A.5 Proposition 7: Financial Solvency Does Not Establish Positive Social Welfare . . . 40
A.6 A Research Checklist . . . 40

List of Figures

1 A schematic representation of an interacting institutional system. The diagram is illustrative and does not assert a specific empirical network. . . . 21

List of Tables

1 Inside–Outside Welfare Classification . . . 10
2 Interdisciplinary Contributions to the General N-Institutions Model . . . 33

1 Introduction

1.1 The General Problem

Institutions organize human activity. Markets coordinate exchange; universities produce and transmit knowledge; courts adjudicate disputes; religious institutions sustain practices and communities; prisons exercise legally authorized coercive powers; central banks influence monetary and financial conditions; and states coordinate public goods, taxation, regulation, diplomacy, and security.

These institutions differ in their objectives, authority, constraints, constituencies, technologies, histories, and relationships with the individuals affected by them. Their outputs cannot automatically be measured in the same units.

Nevertheless, they share a fundamental analytical question:

> How should an institution be evaluated when its activities affect the welfare, behaviour, opportunities, rights, and resources of individuals both inside and outside its organizational boundaries?

Institutional economics studies the rules and organizational structures that shape economic and social interactions [8, 16, 9]. Behavioural economics investigates how psychological, cognitive, social, and contextual factors influence decision-making [6, 14]. Welfare economics asks how alternative states of affairs should be evaluated, including how individual interests may be aggregated [1, 13].

Causal inference provides tools for distinguishing observed associations from the effects of interventions [5, 10]. Financial valuation supplies methods for estimating the present value of cash flows and contingent claims [4]. These fields are complementary, but they do not answer identical questions.

The General N-Institutions Model is intended to coordinate their contributions without conflating their objects of analysis.

1.2 The Central Ordering Principle

The proposed analytical sequence is

$$\boxed{\text{Behaviour} \longrightarrow \text{Causal Inference} \longrightarrow \text{Welfare Effects} \longrightarrow \text{Classification} \longrightarrow \text{Purpose and Valuation.}} \tag{1}$$

This sequence is a methodological proposal, not a theorem claiming that every empirical investigation must follow a rigid chronological order. In practice, classification, measurement, and causal analysis will often proceed iteratively.

The ordering establishes a priority of justification: conclusions about an institution's welfare effects should not be inferred merely from its name, stated mission, or observed popularity.

1.3 The Typology-First Requirement

Let I denote an institution and let $\tau(I)$ denote its organizational type.

Examples of types include:

- market and commercial institutions;
- educational and research institutions;
- religious and cultural institutions;
- judicial and correctional institutions;
- monetary and financial institutions;
- governmental and administrative institutions;
- military and security institutions;
- digital platforms and algorithmic institutions;
- hybrid institutions combining multiple functions.

The type is analytically relevant because it determines which institution-specific outcomes, constraints, legal obligations, stakeholders, and valuation methods must be considered.

However, type alone does not determine welfare effects. A university may expand knowledge while creating financial barriers to access. A market may create mutually beneficial exchanges while imposing external costs on third parties. A prison may impose severe costs on incarcerated individuals while being justified by certain public objectives; whether those objectives are achieved, and whether the means are lawful and proportionate, requires separate investigation.

Consequently, the proposed principle is:

> Classify an institution's observed welfare effects before treating its organizational label or declared purpose as evidence of its social value.

The typology-first requirement means that the analyst first specifies what kind of institution is being examined and which outcomes are appropriate to that type. The welfare-first requirement means that the classification itself depends on evidence rather than on the institution's label.

These requirements are complementary rather than contradictory.

1.4 Scope and Epistemic Status

The treatise contains three kinds of statements.

1. **Formal results:** mathematical conclusions that follow from stated definitions and assumptions.
2. **Methodological principles:** proposed rules for organizing institutional analysis and preventing invalid inference.
3. Empirical hypotheses: claims whose validity depends on observations, measurement, identification assumptions, and replication.

The distinction is essential. A classification rule can be logically consistent without being empirically informative. A statistical association can be precise without identifying a causal effect. A welfare index can be computable without being ethically justified.

The General N-Institutions Model is therefore presented as a research framework requiring empirical validation, not as an already established universal empirical law.

2 Definitions and Basic Notation

2.1 Institutions and Institutional Systems

Let the set of institutions under study be

$$\mathcal{I} = \{I_1, I_2, \ldots, I_N\}, \qquad N \geq 1. \tag{2}$$

Each institution is represented by a profile

$$\Theta_i = (\tau_i, \mathcal{G}_i, \mathcal{R}_i, \mathcal{A}_i, \mathcal{S}_i, \mathcal{T}_i, \mathcal{E}_i)\,, \tag{3}$$

where:

$$\tau_i = \text{organizational type}, \tag{4}$$
$$\mathcal{G}_i = \text{declared and inferred objectives}, \tag{5}$$
$$\mathcal{R}_i = \text{rules, constraints, and procedures}, \tag{6}$$
$$\mathcal{A}_i = \text{authority and decision structure}, \tag{7}$$
$$\mathcal{S}_i = \text{stakeholders and affected populations}, \tag{8}$$
$$\mathcal{T}_i = \text{technologies and productive capabilities}, \tag{9}$$
$$\mathcal{E}_i = \text{external environment}. \tag{10}$$

This profile separates organizational identity from observed performance. It also permits an institution to have several objectives or functions.

An institution may be formally classified as a market institution while performing public functions. A state-owned enterprise may simultaneously exhibit commercial, political, and strategic characteristics. The classification system must therefore permit multiple labels where the evidence warrants them.

2.2 Individuals and Exposure

Let $\mathcal{P}$ denote the relevant population of individuals. For each institution I_i, define an exposure indicator

$$D_{ji} = \begin{cases} 1, & \text{if individual } j \text{ is inside } I_i, \\ 0, & \text{if individual } j \text{ is outside } I_i. \end{cases} \tag{11}$$

This binary representation is a simplifying convention. In practice, membership, participation, authority, dependence, and exposure may be continuous or multidimensional.

An individual may be an employee, customer, student, prisoner, administrator, taxpayer, investor, citizen, neighbour, supplier, or indirectly affected third party. The same person may occupy several roles.

Define the stakeholder-role vector

$$\mathbf{r}_{ji} = (r_{ji1}, r_{ji2}, \ldots, r_{jiK}), \tag{12}$$

where each component represents a relevant relationship with the institution.

A sound analysis should not assume that organizational membership exhausts the set of affected individuals.

2.3 Time, State, and Counterfactual

Let t denote time and T the evaluation horizon.

Let S_{jt} denote the state of individual j, including relevant material, psychological, social, legal, health, and environmental conditions.

Let A_{jt} denote the individual's observed action at time t.

Let I_t denote the institution's state, policies, and activities at time t.

The institutional environment is represented by

$$X_t = (I_t, I_{-i,t}, Z_t)\,, \tag{13}$$

where $I_{-i,t}$ represents other institutions and Z_t contains relevant external conditions.

The counterfactual is a specified alternative institutional regime. It need not be the absence of the institution. It may instead be a different policy, governance structure, technology, organizational form, or institutional arrangement.

2.4 The Institutional Unit of Analysis

The unit of analysis must be explicitly stated. Possible units include:

- the entire institution;
- an organizational division;
- a policy or programme;
- an institutional rule;
- an individual-level interaction;
- a network of institutions;
- an institution over a specified historical interval.

Changing the unit of analysis may change both the estimated effect and the appropriate classification.

For example, an educational institution may have positive aggregate effects while a particular admissions rule creates harmful distributional consequences. A national monetary institution may benefit one group of borrowers while imposing costs on another.

These are not necessarily contradictory findings. They may describe different interventions, populations, or estimands.

3 Behaviour as an Empirical Entry Point

3.1 Behavioural Records

The proposed framework uses observable behaviour as an important source of evidence about institutional functioning and human welfare.

For individual j, define the behavioural record over horizon T as

$$\mathbf{B}_j(T) = (S_{j0}, A_{j0}, S_{j1}, A_{j1}, \ldots, S_{jT}, A_{jT})\,. \tag{14}$$

Behavioural records may include:

- participation and nonparticipation;
- entry, exit, and attempted exit;
- purchases, sales, saving, and investment;
- educational choices and learning activities;

- compliance, resistance, and appeals;
- voting, communication, cooperation, and conflict;
- health-related behaviour;
- use of digital systems;
- responses to incentives, sanctions, and institutional rules.

Behavioural observation can reveal adaptation, strategic response, preference heterogeneity, institutional friction, and changes in opportunity sets.

It does not, by itself, establish whether the observed outcome is beneficial.

3.2 Behavioural Policies

A behavioural policy describes the conditional distribution of actions given a state and institutional environment.

For individual j, define

$$\pi_j(a \mid s, I_t) = \Pr(A_{jt} = a \mid S_{jt} = s, I_t). \tag{15}$$

The policy may be deterministic or stochastic. It may change over time as individuals learn, adapt, experience losses, or respond to social expectations.

A more general behavioural model includes beliefs, constraints, social norms, and latent psychological variables:

$$\pi_j\left(a \mid s, I_t, b_j, c_j, n_j, h_j\right), \tag{16}$$

where:

$$b_j = \text{beliefs and expectations}, \tag{17}$$
$$c_j = \text{constraints and available choices}, \tag{18}$$
$$n_j = \text{social norms and interpersonal influences}, \tag{19}$$
$$h_j = \text{history and prior experience}. \tag{20}$$

These variables are not necessarily directly observable. Their measurement may require surveys, experiments, longitudinal records, or explicit behavioural assumptions.

3.3 Why Observed Behaviour Does Not Identify Welfare

Suppose an individual remains inside an institution. Several explanations are possible:

1. the individual receives substantial benefits;
2. the individual lacks affordable alternatives;
3. exit is prohibited or legally restricted;
4. exit entails financial, social, or psychological costs;
5. the individual expects future benefits despite current losses;
6. the individual is adapting to a constrained environment;
7. the individual's behaviour reflects habit or incomplete information.

The observation is compatible with multiple underlying mechanisms.

The same problem applies to exit. Leaving an institution may indicate harm, but it may also reflect successful completion, a change in preferences, relocation, or an attractive alternative.

Therefore, the following implication is not generally valid:

$$\text{Observed participation} \not\Rightarrow \text{Positive welfare effect}. \tag{21}$$

Nor is the reverse implication valid:

$$\text{Observed nonparticipation} \not\Rightarrow \text{Negative welfare effect.} \tag{22}$$

Behavioural economics and welfare economics jointly motivate this caution. Choices may be affected by bounded rationality, framing, social pressures, and the consequences of the act of choosing itself [6, 14, 2].

3.4 Behaviour, Mechanism, and Welfare

The model distinguishes three objects:

$$\boxed{\text{Behavioural Model} \neq \text{Causal Model} \neq \text{Welfare Model.}} \tag{23}$$

The behavioural model describes actions and transitions.
The causal model estimates how outcomes change under interventions.
The welfare model specifies how the resulting outcomes should be evaluated.
The models can exchange information, but none substitutes for the others.

4 The Welfare-First Classification Principle

4.1 Potential Outcomes

For each individual j and institution I_i, let

$$W_{ji}(1;T) \tag{24}$$

denote the individual's welfare over horizon T under the specified institutional regime.
Let

$$W_{ji}(0;T) \tag{25}$$

denote the corresponding potential welfare under the specified counterfactual.
The individual-level welfare effect is

$$\Delta W_{ji}(T) = W_{ji}(1;T) - W_{ji}(0;T). \tag{26}$$

This equation is meaningful only after the analyst defines:

- the population and unit of analysis;
- the treatment or institutional intervention;
- the counterfactual regime;
- the evaluation horizon;
- the welfare construct and its measurement;
- the relevant assumptions for causal identification.

The potential-outcomes notation follows the general logic of counterfactual causal inference [12, 5].

4.2 Three-Way Classification

For a specified welfare measure, define

$$C_{ji}(T) = \begin{cases} +1, & \Delta W_{ji}(T) > \varepsilon_j, \\ 0, & -\varepsilon_j \leq \Delta W_{ji}(T) \leq \varepsilon_j, \\ -1, & \Delta W_{ji}(T) < -\varepsilon_j, \end{cases} \tag{27}$$

where $\varepsilon_j \geq 0$ is a prespecified tolerance or practical-equivalence threshold.
The three classifications are:

Increasing: the institution produces a sufficiently positive welfare effect relative to the counterfactual.

Maintaining: the effect falls within a prespecified equivalence interval.

Decreasing: the institution produces a sufficiently negative welfare effect relative to the counterfactual.

If the welfare measure is known only imprecisely, classification should incorporate uncertainty rather than assigning a definitive label to every estimated point value.

The tolerance parameter prevents negligible numerical differences from being treated as substantively important. Its value is a methodological choice and must not be selected opportunistically after observing the results.

4.3 Inside and Outside Populations

Define the inside population

$$\mathcal{P}_i^{\text{in}} = \{j \in \mathcal{P} : D_{ji} = 1\}, \tag{28}$$

and the outside population

$$\mathcal{P}_i^{\text{out}} = \{j \in \mathcal{P} : D_{ji} = 0\}. \tag{29}$$

For each group, define a distribution of individual welfare effects:

$$\mathcal{D}_i^{\text{in}} = \left\{\Delta W_{ji}(T) : j \in \mathcal{P}_i^{\text{in}}\right\}, \tag{30}$$

$$\mathcal{D}_i^{\text{out}} = \left\{\Delta W_{ji}(T) : j \in \mathcal{P}_i^{\text{out}}\right\}. \tag{31}$$

These distributions, rather than a single aggregate alone, are the primary inputs to the proposed classification.

Define group-level summaries

$$\mu_i^{\text{in}} = \mathbb{E}\left[\Delta W_{ji}(T) \mid D_{ji} = 1\right], \tag{32}$$

$$\mu_i^{\text{out}} = \mathbb{E}\left[\Delta W_{ji}(T) \mid D_{ji} = 0\right]. \tag{33}$$

These means are useful descriptive statistics, but they do not exhaust the distributional information.

The analyst should also report quantiles, adverse-effect rates, heterogeneity, uncertainty, and outcomes for groups whose interests may be obscured by the mean.

4.4 The Nine-Cell Classification

The initial group-level classification contains nine combinations.

Table 1: Inside–Outside Welfare Classification

Inside effect	Outside effect	Preliminary interpretation
Increasing	Increasing	Broadly beneficial profile
Increasing	Maintaining	Inside benefit without detected outside change
Increasing	Decreasing	Inside benefit with outside harm
Maintaining	Increasing	Outside benefit without detected inside change
Maintaining	Maintaining	No detected material change in either group
Maintaining	Decreasing	Outside harm with no detected inside change
Decreasing	Increasing	Inside harm with outside benefit
Decreasing	Maintaining	Inside harm without detected outside change
Decreasing	Decreasing	Broadly harmful profile

The table is a classification device, not a moral ranking.

In particular, the category "broadly beneficial" does not imply that every individual benefits. Likewise, a positive group mean does not exclude severe losses among a minority.

The classifications should be reported at multiple levels of disaggregation whenever data and privacy constraints permit.

4.5 Proposition 1: Exhaustiveness of the Three-Way Rule

Proposition. For every real-valued welfare effect ΔW and every $\varepsilon \geq 0$, the three-way classification assigns exactly one category.

Proof. Every real number belongs to exactly one of the following mutually exclusive sets:

$$A_+ = (\varepsilon, \infty), \tag{34}$$
$$A_0 = [-\varepsilon, \varepsilon], \tag{35}$$
$$A_- = (-\infty, -\varepsilon). \tag{36}$$

Their union is the real line and their pairwise intersections are empty. Therefore, every $\Delta W \in \mathbb{R}$ belongs to exactly one classification category. □

This is a formal property of the classification rule. It does not establish that welfare has been measured correctly or that the classification is causally identified.

4.6 Proposition 2: Non-Identification of Individual Benefit from Aggregate Benefit

Proposition. A positive average welfare effect does not imply that every individual experiences a positive welfare effect.

Proof. Consider two individuals with welfare effects

$$\Delta W_1 = 4, \qquad \Delta W_2 = -1. \tag{37}$$

The average effect is

$$\overline{\Delta W} = \frac{4 + (-1)}{2} = \frac{3}{2} > 0. \tag{38}$$

Nevertheless, the second individual experiences a negative effect. Hence a positive average does not imply universal individual benefit. □

The proposition motivates distributional reporting and rules against concealing adverse outcomes through aggregation.

4.7 Welfare-First Institutional Classification Principle

The proposed principle is as follows:

> An institution should be classified by its empirically investigated effects on individual welfare inside and outside its boundaries, relative to an explicit counterfactual and over a specified horizon. Its name, declared purpose, and organizational type are relevant contextual information, but they do not independently establish its welfare effects.

This principle does not imply that welfare effects alone reveal an institution's purpose. Different objectives, technologies, legal arrangements, and historical processes can generate similar observed outcomes.

Purpose, legitimacy, rights, and welfare must therefore be examined as distinct but interacting dimensions.

5 Defining and Measuring Welfare

5.1 Welfare Is a Construct, Not a Raw Observation

Individual welfare is not directly observed in the same way as a transaction price or an account balance. It is inferred from theoretically and empirically justified indicators.

Let the welfare-relevant vector for individual j be

$$\mathbf{Y}_{jt} = \left(Y_{jt}^m, Y_{jt}^h, Y_{jt}^p, Y_{jt}^c, Y_{jt}^r, Y_{jt}^s, Y_{jt}^e\right), \tag{39}$$

where the components may represent:

$$Y^m = \text{material conditions}, \tag{40}$$
$$Y^h = \text{physical health}, \tag{41}$$
$$Y^p = \text{psychological well-being}, \tag{42}$$
$$Y^c = \text{capabilities and opportunities}, \tag{43}$$
$$Y^r = \text{rights and security}, \tag{44}$$
$$Y^s = \text{social relations and participation}, \tag{45}$$
$$Y^e = \text{environmental conditions}. \tag{46}$$

The vector is illustrative rather than exhaustive. The dimensions must be adapted to the institutional context and the population under investigation.

The capability approach emphasizes substantive freedoms and the opportunities people actually possess, rather than resources or subjective satisfaction alone [13, 11].

This is especially important when formal access does not translate into effective access because of disability, poverty, discrimination, coercion, infrastructure, or other conversion factors.

5.2 A Multidimensional Welfare Representation

A general individual-level welfare representation is

$$W_j = F_j\left(\mathbf{Y}_j, \mathbf{K}_j, \mathbf{Q}_j\right), \tag{47}$$

where:

$$\mathbf{Y}_j = \text{observed or estimated outcomes}, \tag{48}$$
$$\mathbf{K}_j = \text{capabilities and effective opportunities}, \tag{49}$$
$$\mathbf{Q}_j = \text{rights, constraints, and relevant context}. \tag{50}$$

The function F_j is not assumed to be known universally. It may be estimated, elicited, specified normatively, or represented by partial-order comparisons.

If the available information does not justify a unique scalar measure, the model should preserve a vector or partial ordering instead of manufacturing a single index.

5.3 Subjective and Objective Indicators

Subjective indicators may include:

- self-reported life satisfaction;
- experienced stress;
- perceived autonomy;
- trust and belonging;
- perceived fairness;
- perceived safety.

Objective indicators may include:

- income and consumption;
- mortality and morbidity;
- educational attainment;
- housing conditions;
- exposure to violence or environmental hazards;

- legal access and procedural outcomes;
- availability of meaningful alternatives.

Neither category is sufficient in isolation for every application. Revealed preferences can misrepresent welfare when preferences depend on choice sets or when choosing itself changes experienced welfare. Subjective happiness and satisfaction are not universally sufficient statistics either [2].

The proposed solution is triangulation: use several theoretically relevant sources of evidence and investigate disagreements between them.

5.4 Non-Compensable Constraints

Some dimensions may be treated as constraints rather than as ordinary additive welfare components.

For example, an analysis may impose minimum requirements involving basic rights, physical safety, due process, or protection from prohibited discrimination.

Let $\mathcal{F}_i$ denote the feasible set of institutionally permissible outcomes. Then the valuation problem may be written as

$$\max_{\mathbf{x}\in\mathcal{F}_i} \mathcal{V}_i(\mathbf{x}) \tag{51}$$

subject to applicable legal and ethical constraints.

A high estimated economic benefit does not automatically make an otherwise impermissible action acceptable. The model must distinguish ordinary trade-offs from constraints that cannot legitimately be traded away under the governing legal and ethical framework.

6 Causal Inference and Counterfactual Identification

6.1 Potential Outcomes

Let $D_{ji} \in \{0, 1\}$ represent the specified institutional exposure. Define potential outcomes

$$W_{ji}(1) = \text{welfare under exposure,} \tag{52}$$
$$W_{ji}(0) = \text{welfare under the counterfactual.} \tag{53}$$

The individual treatment effect is

$$\tau_{ji} = W_{ji}(1) - W_{ji}(0). \tag{54}$$

The fundamental identification problem is that, for the same individual and the same time horizon, only one potential outcome is ordinarily observed under a particular realized regime.

The unobserved potential outcome cannot simply be replaced by an assumption that the individual would have experienced zero welfare change without the institution.

6.2 Target Estimands

The average treatment effect is

$$\text{ATE}_i = \mathbb{E}\left[W_i(1) - W_i(0)\right]. \tag{55}$$

The average treatment effect among exposed individuals is

$$\text{ATT}_i = \mathbb{E}\left[W_i(1) - W_i(0) \mid D_i = 1\right]. \tag{56}$$

These estimands answer different questions. The ATE concerns the target population under the stated intervention. The ATT concerns those actually exposed.

Neither necessarily equals the effect on individuals who were excluded, who left, or who never had a realistic opportunity to participate.

6.3 Identification Assumptions

A conventional observational identification strategy may require the following assumptions.

Consistency. The observed outcome equals the potential outcome corresponding to the realized treatment regime.

$$W_i = D_i W_i(1) + (1 - D_i) W_i(0). \tag{57}$$

Conditional exchangeability. Given measured pre-treatment covariates X_i, exposure is independent of the potential outcomes:

$$(W_i(1), W_i(0)) \perp D_i \mid X_i. \tag{58}$$

Positivity. Every relevant covariate profile has a nonzero probability of receiving each treatment regime:

$$0 < \Pr(D_i = 1 \mid X_i = x) < 1. \tag{59}$$

These assumptions are substantive, not merely algebraic. In particular, conditional exchangeability is not guaranteed by fitting a regression model.

The appropriate identification strategy depends on the intervention, data-generating process, ethical constraints, and institutional environment [5, 10].

6.4 The G-Computation Formula

Under consistency, conditional exchangeability, and positivity, the population mean under treatment can be identified as

$$\mathbb{E}[W(1)] = \mathbb{E}_X \left[\mathbb{E}[W \mid D = 1, X] \right]. \tag{60}$$

Similarly,

$$\mathbb{E}[W(0)] = \mathbb{E}_X \left[\mathbb{E}[W \mid D = 0, X] \right]. \tag{61}$$

Subtracting gives

$$\begin{aligned} \text{ATE} = \mathbb{E}_X \left[\mathbb{E}[W \mid D = 1, X] \right] \\ - \mathbb{E}_X \left[\mathbb{E}[W \mid D = 0, X] \right]. \end{aligned} \tag{62}$$

The formula identifies the estimand only if the assumptions are defensible for the specific institutional setting.

6.5 Difference-in-Differences

Where treatment timing and comparison groups permit a difference-in-differences design, a basic two-period estimand is

$$\begin{aligned} \widehat{\delta}_{\text{DiD}} = \left(\overline{W}_{T,1} - \overline{W}_{T,0} \right) \\ - \left(\overline{W}_{C,1} - \overline{W}_{C,0} \right), \end{aligned} \tag{63}$$

where T denotes the treated group and C the comparison group.

A causal interpretation requires an appropriate parallel-trends assumption and attention to treatment timing, composition changes, anticipation, spillovers, and concurrent interventions.

This estimator is not automatically valid merely because outcomes are available before and after a policy change.

6.6 Selection, Exit, and Survivorship

A particularly important institutional problem is selective participation.

Let $R_{ji} = 1$ indicate that the individual remains observable after institutional exposure. If continued observation depends on welfare or treatment effects, estimates based only on observed participants may be biased.

For example, a harmful institution may lose its most dissatisfied participants. The remaining population can appear relatively satisfied because those who experienced the worst outcomes have left.

The model therefore distinguishes:

- current participants;
- former participants;
- applicants who were rejected;
- individuals who could not afford to enter;
- individuals who declined entry;
- individuals indirectly affected without formal participation.

Where ethically and legally permissible, longitudinal follow-up should include these groups.

6.7 Spillovers and Interference

Institutions frequently affect individuals who are not formally exposed. This violates the simplest no-interference assumptions.

Let G_{ji} represent the exposure of individual j to institutional spillovers, such as network effects, externalities, price changes, social influence, or changes in public security.

Potential outcomes may then be written as

$$W_{ji} = W_{ji}(D_{ji}, G_{ji}). \tag{64}$$

The total effect of an institutional intervention may include:

- direct effects on participants;
- indirect effects on nonparticipants;
- equilibrium effects through prices or wages;
- network effects through other institutions;
- long-run effects through learning and adaptation.

An analysis that ignores these channels may underestimate or misattribute institutional effects.

7 Econometric Measurement and Statistical Uncertainty

7.1 A General Panel Specification

Suppose the analyst observes individuals j, institutions i, and periods t. A baseline specification is

$$\begin{aligned} Y_{jit} = \alpha_j + \lambda_t + \beta D_{jit} + \boldsymbol{\gamma}^{\mathsf{T}} \mathbf{X}_{jit} \\ + \boldsymbol{\eta}^{\mathsf{T}} \mathbf{Z}_{it} + \varepsilon_{jit}, \end{aligned} \tag{65}$$

where:

$$\alpha_j = \text{individual fixed effect}, \tag{66}$$

$$\lambda_t = \text{time effect}, \tag{67}$$

$$D_{jit} = \text{institutional exposure}, \tag{68}$$

$$\mathbf{X}_{jit} = \text{individual-level covariates}, \tag{69}$$

$$\mathbf{Z}_{it} = \text{institution-level covariates}, \tag{70}$$

$$\varepsilon_{jit} = \text{unobserved component}. \tag{71}$$

The coefficient β has a causal interpretation only under the assumptions appropriate to the chosen design. Fixed effects alone do not eliminate time-varying confounding or solve all selection problems.

Standard errors must reflect the sampling design and dependence structure. Where observations are clustered within institutions, naive independent-observation standard errors can substantially misrepresent uncertainty.

7.2 Measurement Models

If welfare is latent, a measurement model may relate observed indicators to an underlying construct.

For indicator k, one possible specification is

$$Y_{jkt} = \nu_k + \lambda_k W_{jt} + e_{jkt}, \tag{72}$$

where λ_k is a factor loading and e_{jkt} is a measurement error.

This representation requires substantive assumptions about the relationship between the latent construct and observed indicators.

Measurement invariance should be investigated across:

- institutional types;
- demographic groups;
- languages and cultural contexts;
- time periods;
- participation and nonparticipation groups.

If the same survey response has different meanings across populations, direct comparisons may be invalid without additional modelling.

7.3 Uncertainty-Aware Classification

Suppose an estimator $\widehat{\tau}_i$ estimates a group-level welfare effect and $\widehat{\sigma}_i$ estimates its standard error.

The confidence interval may be written as

$$\mathcal{C}_i = \left[\widehat{\tau}_i - z_{1-\alpha/2}\widehat{\sigma}_i,\ \widehat{\tau}_i + z_{1-\alpha/2}\widehat{\sigma}_i\right]. \tag{73}$$

A conservative classification rule is:

- classify as increasing only when the lower confidence bound exceeds the positive threshold;
- classify as decreasing only when the upper confidence bound falls below the negative threshold;
- classify as maintaining only when an appropriate equivalence procedure supports that conclusion;
- otherwise report the result as inconclusive.

Failure to reject a zero-effect null hypothesis does not establish that the effect is zero. Equivalence requires a prespecified equivalence margin and an appropriate inferential procedure.

7.4 Robustness and Sensitivity Analysis

Every empirical classification should examine:

1. alternative counterfactuals;
2. alternative welfare indicators;
3. alternative time horizons;
4. heterogeneous treatment effects;
5. omitted-variable sensitivity;
6. measurement error;

7. missing-data assumptions;
8. alternative model specifications;
9. possible spillovers and equilibrium effects;
10. the influence of extreme observations.

A classification that changes under small defensible specification changes should be reported as fragile.

The objective is not to maximize the number of institutions assigned a definitive label. It is to identify which conclusions are supported by the evidence and which remain uncertain.

8 Behavioural Economics, Psychology, and Learning

8.1 Bounded Rationality

The General N-Institutions Model does not require every individual to maximize a stable, fully informed utility function.

Decision-making may be constrained by:

- limited attention;
- imperfect information;
- computational limitations;
- habit and learning;
- reference dependence;
- social norms;
- framing and defaults;
- loss aversion and present bias;
- stress and environmental pressure.

These mechanisms can explain why observed choices may differ from the choices predicted by a simple rational-choice model [6, 14].

They do not, by themselves, determine the correct welfare criterion. A behavioural deviation from a standard economic model is not automatically evidence of irrationality or harm.

8.2 Institutional Feedback

Institutions change the states in which individuals make decisions. Individuals, in turn, respond to institutional rules and can change the institution through collective action.

A general state-transition representation is

$$S_{j,t+1} \sim P_j \left(S_{j,t+1} \mid S_{jt}, A_{jt}, I_t, \mathbf{X}_t \right). \tag{74}$$

Institutional dynamics may be represented by

$$I_{t+1} = \Phi \left(I_t, \mathbf{A}_t, \mathbf{S}_t, \mathbf{Z}_t \right). \tag{75}$$

The first equation describes how individual states evolve. The second describes how institutional rules, policies, or structures evolve in response to behaviour and external conditions.

These equations define a possible modelling architecture. Their functional forms and causal interpretation must be established empirically for each application.

8.3 Learning and Adaptation

An institution may improve welfare by increasing knowledge, skills, or access to information. It may also produce dependence, learned helplessness, maladaptive habits, or excessive reliance on automated systems.

A model should therefore distinguish:

- immediate effects;
- learning effects;
- adaptation effects;
- habit formation;
- persistent changes in capabilities;
- intergenerational transmission.

The sign of a short-run effect need not match the sign of a long-run effect.

A dynamic welfare measure can be defined as

$$\mathcal{W}_j(T) = \sum_{t=0}^{T} \delta_j^t u_j \left(\mathbf{Y}_{jt}, \mathbf{K}_{jt} \right), \tag{76}$$

where δ_j is a specified discount factor and u_j is a welfare representation.

The discount factor and the welfare function require justification. A financial discount rate should not automatically be used as the ethical discount rate for human welfare.

8.4 Psychiatric and Psychological Evidence

Where an institution affects mental health, relevant outcomes may include stress, anxiety, social isolation, perceived control, psychological safety, and access to appropriate care.

Such outcomes require valid measurement and, where necessary, assessment by qualified professionals.

The model must not infer a psychiatric diagnosis from ordinary behavioural data alone. Nor should it equate institutional compliance with psychological health.

Data involving mental health require heightened privacy protections, purpose limitation, and careful consideration of the risks of stigmatization or discriminatory use.

9 Institutional Purpose, Legitimacy, and Law

9.1 Purpose Is Not Identical to Effect

Let $\mathcal{G}_i$ represent the institution's objectives. These may be stated in statutes, charters, contracts, policy documents, public declarations, or internal operating procedures.

Let $\mathcal{O}_i$ represent observed outcomes.

There is no general logical implication

$$\mathcal{G}_i \not\Rightarrow \mathcal{O}_i. \tag{77}$$

An institution may fail to achieve its declared objectives. It may achieve them only for selected groups. Its stated purpose may also differ from the incentives governing actual decisions.

Consequently, purpose should be investigated using documentary evidence, governance records, incentives, decision procedures, historical development, and observed conduct.

Behavioural evidence can help explain institutional operation, but it does not uniquely identify the institution's true purpose.

9.2 Legitimacy as a Distinct Dimension

The proposed framework distinguishes:

1. welfare consequences;
2. legal validity;
3. procedural legitimacy;
4. democratic accountability;
5. rights compliance;
6. effectiveness in achieving declared objectives;
7. financial sustainability.

These dimensions can conflict.

A policy might produce an aggregate economic gain while violating applicable rights. An institution might operate legally while performing poorly. An institution might be financially sustainable but distribute benefits and burdens in ways that raise serious ethical objections.

The analytical task is to expose such conflicts rather than compressing them into an unqualified label.

9.3 Rights and Human-Rights Due Diligence

For business institutions, the United Nations Guiding Principles on Business and Human Rights provide a framework organized around the state duty to protect, the corporate responsibility to respect, and access to remedy [15].

The broader lesson for institutional analysis is that adverse effects require identification, prevention or mitigation where applicable, accountability, and access to remedy.

This does not mean that every institution is governed by precisely the same legal obligations. Duties vary by jurisdiction, legal form, function, treaty obligations, and the rights and interests involved.

The model therefore includes a legal mapping module rather than assuming a universal substantive code.

9.4 Legal Classification

Let $\mathcal{L}_i$ denote the applicable legal framework for institution I_i.

The legal assessment can be represented as a vector

$$\mathbf{L}_i = \left(L_i^{\text{authority}}, L_i^{\text{procedure}}, L_i^{\text{rights}}, L_i^{\text{liability}}, L_i^{\text{remedy}} \right). \tag{78}$$

The components describe the legal basis of authority, procedural requirements, rights constraints, liability, and available remedies.

This is a classification architecture, not a substitute for jurisdiction-specific legal analysis.

10 Social Choice and Aggregation

10.1 The Aggregation Problem

Individual welfare effects do not automatically generate a unique social welfare ranking.

Suppose the population has J individuals. A utilitarian aggregation rule may be written as

$$SW = \sum_{j=1}^{J} \omega_j W_j, \tag{79}$$

where ω_j denotes a social weight.

The weights are not supplied by mathematics alone. Their selection requires normative justification and sensitivity analysis.

Moreover, the formula presupposes that the individual welfare measures can be meaningfully compared or transformed to a common scale. This is a substantive assumption.

10.2 Distributional Welfare

A distribution-sensitive social evaluation can be written as

$$SW = \mathcal{S}(W_1, W_2, \ldots, W_J), \tag{80}$$

where $\mathcal{S}$ is an explicitly chosen social evaluation function.

Possible approaches include:

- utilitarian aggregation;
- prioritarian weighting;
- sufficientarian thresholds;
- capability-based evaluation;
- rights-constrained optimization;
- partial-order comparisons;
- multi-criteria decision analysis.

These approaches embody different ethical commitments. The model should report the chosen criterion rather than describing it as value-neutral.

10.3 Compensation and Pareto Comparisons

A Pareto improvement occurs when at least one individual becomes better off and no individual becomes worse off, under the specified welfare representation.

A potential-compensation criterion asks whether gains could in principle compensate losses. It does not establish that compensation actually occurs or that the distribution is ethically acceptable.

The distinction is crucial when an institution creates concentrated losses and diffuse benefits.

10.4 Arrow's Impossibility Result

Social choice theory establishes important limitations on the construction of collective preference orderings. Arrow's theorem shows that, for more than two alternatives, no social welfare function satisfies all of a particular set of conditions, including unrestricted domain, social ordering, Pareto responsiveness, independence of irrelevant alternatives, and non-dictatorship [1].

The result does not prohibit all useful social evaluation. It shows that certain desirable requirements cannot all be maintained simultaneously in the specified framework.

For the General N-Institutions Model, the implication is methodological: a universal scalar ranking cannot be presumed to satisfy every reasonable criterion of fairness, autonomy, consistency, and collective choice.

Where a scalar score is used, its assumptions and trade-offs must be explicit.

10.5 A Distributional Reporting Vector

For institution I_i, define

$$\mathbf{D}_i = \left(\mu_i, q_{i,0.1}, q_{i,0.5}, q_{i,0.9}, p_i^-, p_i^+, h_i\right), \tag{81}$$

where:

$$\mu_i = \text{mean welfare effect}, \tag{82}$$

$$q_{i,p} = p\text{-quantile of welfare effects}, \tag{83}$$

$$p_i^- = \text{share experiencing material harm}, \tag{84}$$

$$p_i^+ = \text{share experiencing material benefit}, \tag{85}$$

$$h_i = \text{selected heterogeneity statistic}. \tag{86}$$

The reporting vector is illustrative. The selected quantiles, thresholds, and heterogeneity measures should fit the application.

It prevents the mean from being the only reported measure of institutional performance.

11 The General N-Institutions System

11.1 Institutional Networks

Institutions interact through financial, legal, political, informational, technological, social, and security relationships.

Represent the system as a directed weighted graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A}), \tag{87}$$

where:

$$\mathcal{V} = \{I_1, \ldots, I_N\}, \tag{88}$$

$$\mathcal{E} = \text{directed institutional relationships}, \tag{89}$$

$$\mathbf{A} = \text{weighted adjacency matrix}. \tag{90}$$

The matrix element a_{ik} represents a specified relationship from institution I_k to institution I_i. Its meaning and units must be defined for the application.

Examples include:

- a central bank influencing commercial banks;
- a legislature authorizing an administrative agency;
- a court reviewing an executive action;
- a university supplying skilled workers to firms;
- a financial market transmitting shocks across jurisdictions;
- a security alliance changing the incentives of member states.

11.2 Graphical Representation

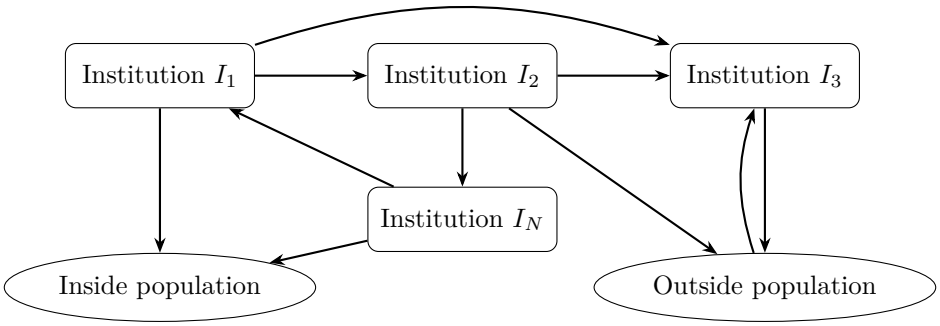


Figure 1: A schematic representation of an interacting institutional system. The diagram is illustrative and does not assert a specific empirical network.

The graph represents relationships, not necessarily causal effects. A directed edge should be interpreted causally only when supported by an appropriate identification strategy.

11.3 Cross-Institutional Exposure

Let $\mathbf{d}_j$ denote the exposure vector of individual j across the N institutions:

$$\mathbf{d}_j = (d_{j1}, d_{j2}, \ldots, d_{jN}). \tag{91}$$

The welfare outcome may depend on the full vector:

$$W_j = W_j\left(d_{j1}, d_{j2}, \ldots, d_{jN}\right). \tag{92}$$

This formulation allows institutional complementarity and substitution. For example, the effect of a university may depend on labour-market conditions, financial access, transport infrastructure, and public health.

11.4 Interaction Effects

For two institutions I_i and I_k, define the interaction effect

$$\begin{aligned}\Gamma_{ik} = W(1,1) - W(1,0) \\ - W(0,1) + W(0,0).\end{aligned} \tag{93}$$

If $\Gamma_{ik} > 0$, the joint effect exceeds the sum of the two individual effects under the specified outcome scale. If $\Gamma_{ik} < 0$, the interaction is subadditive.

The interpretation depends on the outcome scale and the intervention definition. Interaction is not necessarily evidence of cooperation or conflict in a moral sense.

For N institutions, higher-order interactions may exist. A model that ignores them may misattribute the effect of a system to one component.

11.5 Institutional Complementarity and Substitution

An institution may:

- complement another institution's capabilities;
- substitute for an institution that is absent or ineffective;
- increase another institution's costs;
- transfer risks to other institutions;
- create dependencies that amplify systemic failure.

The analysis should therefore consider both institution-specific effects and system-level effects.

A positive effect for one institution does not establish a positive effect for the institutional system as a whole.

12 A Formal General N-Institutions Model

12.1 Definition

The proposed General N-Institutions Model is the tuple

$$\mathfrak{M}_N = (\mathcal{I}, \Theta, \mathcal{P}, \mathbf{B}, \mathbf{Y}, \mathbf{W}, \mathcal{C}, \mathcal{G}, \mathcal{L}, \mathcal{V}), \tag{94}$$

where:

$$\mathcal{I} = \text{set of institutions}, \tag{95}$$
$$\Theta = \text{institutional profiles}, \tag{96}$$
$$\mathcal{P} = \text{relevant population}, \tag{97}$$
$$\mathbf{B} = \text{behavioural records}, \tag{98}$$
$$\mathbf{Y} = \text{observed outcome vectors}, \tag{99}$$
$$\mathbf{W} = \text{welfare representations}, \tag{100}$$
$$\mathcal{C} = \text{counterfactual and causal specifications}, \tag{101}$$
$$\mathcal{G} = \text{institutional interaction network}, \tag{102}$$
$$\mathcal{L} = \text{legal and ethical constraints}, \tag{103}$$
$$\mathcal{V} = \text{institution-specific valuation functions}. \tag{104}$$

The tuple is a proposed formal architecture. It does not claim that all components can be measured perfectly or that one estimation method is appropriate for every institution.

12.2 A Hierarchy of Questions

For every institution I_i, the model asks:

1. What is the institutional unit of analysis?
2. What organizational type or types apply?
3. Who participates, who is excluded, and who is indirectly affected?
4. What behaviours and outcomes can be observed?
5. What causal intervention and counterfactual are relevant?
6. Which welfare dimensions are appropriate?
7. What distribution of welfare effects is estimated?
8. What uncertainties and identification limitations remain?
9. What purpose, legal obligations, and legitimacy criteria apply?
10. Which valuation methods are justified by the institution's objectives and constraints?

The questions establish an auditable route from observation to valuation.

12.3 Proposition 3: Conditional Comparability

Proposition. Two institutions can be compared on a common scalar criterion only when a common comparison rule has been specified and is applicable to both institutions.

Proof. A scalar comparison requires a mapping from each institution's relevant outcomes into a common ordered set. Let these mappings be V_i and V_k. A common scalar ranking requires that their outputs have a common interpretation and that the comparison rule is defined over both output sets.

If the mappings have incompatible units or meanings, the expression $V_i - V_k$ is not intrinsically meaningful. A common mapping or an explicit transformation is therefore a prerequisite for scalar comparison. □

The proposition is conditional and methodological. It does not prove that institutions can never be compared. It establishes that comparability requires an explicit mapping rather than an unexamined assumption.

12.4 Institutional Non-Equivalence Principle

The proposed principle is:

> A common analytical framework does not imply substantive equivalence between the institutions it analyzes. Comparison requires a justified mapping between the relevant outcomes, constraints, objectives, and valuation criteria.

This principle allows a common procedure without forcing a market, university, religious institution, prison, central bank, and state to share a single intrinsic-value definition.

13 Institutional Valuation

13.1 Valuation Is Not a Single Universal Quantity

The term "value" may refer to:

- market price;
- financial enterprise value;
- asset value;
- fiscal contribution;

- operational effectiveness;
- social welfare effects;
- public value;
- legal or constitutional significance;
- strategic or security value;
- resilience and option value.

These concepts are not interchangeable.
The proposed valuation object is a vector

$$\mathbf{V}_i = \left(V_i^F, V_i^O, V_i^W, V_i^R, V_i^S, V_i^G\right), \tag{105}$$

where:

$$V_i^F = \text{financial value}, \tag{106}$$
$$V_i^O = \text{operational value}, \tag{107}$$
$$V_i^W = \text{welfare effects}, \tag{108}$$
$$V_i^R = \text{rights and legitimacy assessment}, \tag{109}$$
$$V_i^S = \text{systemic contribution and risk}, \tag{110}$$
$$V_i^G = \text{governance and institutional quality}. \tag{111}$$

The vector can be extended or simplified according to the application.
The components should not be added together unless the analyst specifies a defensible aggregation rule.

13.2 Financial Valuation

For an institution with forecastable cash flows, discounted cash-flow valuation may be appropriate.

Let FCF_t denote free cash flow in period t, and let r_t denote the applicable discount rate. A simple present-value model is

$$V_0^F = \sum_{t=1}^{T} \frac{FCF_t}{(1+r_t)^t} + \frac{TV_T}{(1+r_T)^T}, \tag{112}$$

where TV_T denotes terminal value.
This method requires assumptions about:

- future cash flows;
- risk and discount rates;
- capital structure;
- taxes and regulation;
- reinvestment;
- terminal growth;
- contingent liabilities;
- the valuation date and horizon.

The method is suitable only when the relevant cash flows and claims can be meaningfully defined. It does not automatically measure the social value of a university, court, temple, prison, or public agency.

13.3 Stochastic Discount Factors

In asset pricing, a stochastic discount factor $M_{t,T}$ can express the price of a payoff X_T as

$$P_t = \mathbb{E}_t \left[M_{t,T} X_T \right]. \tag{113}$$

This is a powerful financial valuation representation [3].

However, the equation prices a payoff under a specified financial pricing framework. It does not independently define the social welfare function of an institution.

The distinction between financial pricing and social evaluation must remain explicit.

13.4 Public Institutions and Social Cost-Benefit Analysis

Where market prices fail to capture relevant externalities, a public evaluation may require social cost-benefit analysis.

Let B_t denote benefits, C_t costs, and r_s the selected social discount rate. A simplified net present social value is

$$NPSV = \sum_{t=0}^{T} \frac{B_t - C_t}{(1 + r_s)^t}. \tag{114}$$

The result depends on how benefits, costs, externalities, distribution, and the social discount rate are specified.

Nonmarket outcomes may require shadow prices or nonmonetary indicators. Where monetization is ethically inappropriate or empirically indefensible, the analysis should retain separate indicators and constraints.

13.5 Financial Planning and Analysis

For institutions with budgets, define

$$\Pi_t = R_t - O_t - C_t, \tag{115}$$

where:

$$R_t = \text{revenue or funding}, \tag{116}$$
$$O_t = \text{operating expenditure}, \tag{117}$$
$$C_t = \text{capital expenditure}. \tag{118}$$

A budget surplus or deficit is an important financial indicator, but it is not identical to institutional effectiveness or social welfare.

A public health institution may operate a deficit while generating substantial public benefits. A profitable institution may generate external costs. Conversely, financial sustainability may be essential for an institution to continue delivering its intended benefits.

The model therefore treats financial sustainability as one component of institutional performance rather than as a universal proxy for value.

13.6 Risk-Adjusted and Scenario-Based Valuation

For uncertain outcomes, valuation may require scenario analysis.

Let $s \in \mathcal{S}$ index scenarios and p_s their assigned probabilities. A financial expectation is

$$\mathbb{E}[V_i] = \sum_{s \in \mathcal{S}} p_s V_i(s), \qquad \sum_{s \in \mathcal{S}} p_s = 1. \tag{119}$$

Scenario probabilities may be estimated, elicited, or used as explicit stress assumptions. They should not be treated as objective frequencies when they are merely subjective judgements.

A robust valuation should also examine tail risks, model uncertainty, liquidity constraints, and the consequences of failure.

14 Physics, Chemistry, Biology, and Complex Systems

14.1 Physics: State, Dynamics, and Stability

Physics contributes a disciplined language for states, dynamics, feedback, stability, and system boundaries.

A general dynamical representation is

$$\mathbf{x}_{t+1} = \mathbf{F}\left(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\epsilon}_t\right), \tag{120}$$

where $\mathbf{x}_t$ is the system state, $\mathbf{u}_t$ is an intervention, and $\boldsymbol{\epsilon}_t$ represents disturbances.

This form is useful for modelling institutional dynamics, but the mathematical analogy does not establish that social systems obey the same substantive laws as physical systems.

Institutional agents learn, reinterpret rules, change objectives, and react to measurement. These properties complicate the direct transfer of physical models.

14.2 Feedback and Stability

Suppose a simplified institutional state follows

$$x_{t+1} = f(x_t). \tag{121}$$

A fixed point x^* satisfies

$$x^* = f(x^*). \tag{122}$$

In a differentiable one-dimensional system, a local condition for stability is

$$|f'(x^*)| < 1. \tag{123}$$

This condition illustrates how feedback may stabilize or destabilize an institutional process.

Its use in an empirical institutional model requires an appropriate state variable, a defensible transition equation, and evidence that the local approximation is meaningful.

14.3 Chemistry: Stocks, Flows, and Conservation

Chemistry provides useful analogies for stocks, flows, reactions, and material constraints. Institutional accounting similarly distinguishes stocks from flows.

For a financial or physical stock K_t, a generic accounting identity may be written as

$$K_{t+1} = K_t + F_t^{\text{in}} - F_t^{\text{out}} + \Delta_t, \tag{124}$$

where Δ_t represents revaluation, depreciation, destruction, or other adjustments.

The identity is useful for resource accounting, but it is not a complete model of institutional welfare.

14.4 Biology: Adaptation and Ecological Interdependence

Biology contributes concepts of adaptation, resilience, ecological interaction, and dependence on environmental conditions.

Institutional systems may exhibit analogous properties:

- adaptation to changing constraints;
- resilience after shocks;
- interdependence among subsystems;
- competition for scarce resources;
- vulnerability to common disturbances.

These analogies must not be used to naturalize political arrangements or imply that existing institutions are morally justified because they persist.

Institutional survival and social benefit are distinct empirical questions.

14.5 Environmental Externalities

Environmental outcomes may be relevant to welfare inside and outside an institution.

Let E_t denote an environmental state and a_t institutional activity. A simplified dynamic is

$$E_{t+1} = g(E_t, a_t, \mathbf{z}_t), \tag{125}$$

where $\mathbf{z}_t$ contains external environmental drivers.

Environmental effects may be delayed, spatially distributed, and intergenerational. Their measurement may require scientific models outside economics, including atmospheric science, ecology, toxicology, and public health.

Interdisciplinary evidence should be integrated with explicit uncertainty rather than reduced to an unsupported numerical score.

15 Political Science, Geopolitics, and Diplomacy

15.1 Political Institutions

Political institutions allocate authority, establish rules, and determine how collective decisions are made.

The model may examine:

- representation and accountability;
- administrative capacity;
- public-goods provision;
- corruption and rent extraction;
- legitimacy and compliance;
- conflict resolution;
- distribution of political power.

The welfare effects of political institutions may vary across citizens, residents, noncitizens, future generations, and individuals outside the jurisdiction.

A jurisdictional boundary is therefore not necessarily the boundary of the institution's relevant welfare effects.

15.2 Geopolitical Spillovers

States and international organizations can affect foreign populations through trade, migration, finance, sanctions, security arrangements, environmental effects, and armed conflict.

For a state institution I_i, the population of interest may be partitioned into domestic and foreign groups:

$$\mathcal{P} = \mathcal{P}_i^{\text{domestic}} \cup \mathcal{P}_i^{\text{foreign}}. \tag{126}$$

This partition is illustrative. A complete analysis may require additional categories for refugees, diasporas, transnational workers, future generations, and other affected populations.

Domestic welfare improvements should not be treated as a complete assessment of a policy whose costs are transferred abroad.

15.3 Diplomacy and Institutional Cooperation

International institutions may produce gains through coordination, information sharing, dispute resolution, and credible commitments.

They may also generate unequal bargaining power, enforcement problems, strategic dependence, and distributional conflict.

A cooperative arrangement should therefore be assessed by:

1. the welfare effects on each participant;

2. the effects on nonparticipants;
3. the distribution of costs and benefits;
4. compliance and enforcement;
5. resilience to defection or external shocks;
6. the legal basis and legitimacy of its decisions.

The framework does not presume that cooperation is always beneficial or that competition is always harmful. Both propositions require context-specific evidence.

15.4 Warfare Economics and Security

Security institutions create a particularly difficult valuation problem because their activities may affect deterrence, survival, civilian harm, public expenditure, civil liberties, and international stability.

A security analysis must distinguish:

- defensive capability;
- deterrence and escalation risk;
- civilian harm;
- opportunity costs of expenditure;
- displacement and long-term damage;
- compliance with applicable international law;
- effects on domestic and foreign populations.

A security benefit cannot be established solely by counting weapons, personnel, expenditure, or territorial control. These are inputs or intermediate indicators, not complete measures of welfare or security.

Likewise, a favourable financial return does not establish the legality or legitimacy of a military action.

The framework is intended for policy evaluation, risk assessment, and institutional accountability, not for treating human suffering as an ordinary financial externality.

16 Computer Science, Data Science, and Algorithms

16.1 The Institutional Data Pipeline

The empirical workflow can be organized as

$$\boxed{\text{Data} \to \text{Validation} \to \text{Measurement} \to \text{Identification} \to \text{Classification} \to \text{Audit.}} \tag{127}$$

The pipeline should retain provenance at every stage.

Data engineering should record:

- source and collection date;
- population coverage;
- missingness and exclusion rules;
- transformations and feature definitions;
- model versions;
- assumptions and estimands;
- uncertainty estimates;
- human review and approval.

The objective is reproducibility and accountability, not merely automation.

16.2 Pseudocode for Institutional Classification

The following pseudocode is deliberately independent of any particular programming language.

```
INPUT:
    institutions I_1, ..., I_N
    relevant populations P
    observed behavioural and outcome data
    institutional profiles
    candidate counterfactuals
    welfare definitions
    legal and ethical constraints
    evaluation horizons

FOR each institution I_i:

    1. Define the unit of analysis.
    2. Identify organizational type and hybrid functions.
    3. Map inside, outside, excluded, and indirectly affected groups.
    4. Record data sources and measurement limitations.
    5. Specify the intervention and counterfactual.
    6. Select an identification strategy.
    7. Estimate outcome effects and uncertainty.
    8. Test assumptions and sensitivity.
    9. Estimate effects separately across relevant groups.
   10. Construct welfare-effect distributions.
   11. Apply prespecified classification thresholds.
   12. Assess rights, legality, and procedural legitimacy.
   13. Estimate institution-specific financial and operational value.
   14. Record unresolved uncertainties and model limitations.
   15. Produce an auditable institutional report.

OUTPUT:
    classification profile
    causal estimates
    welfare-effect distributions
    legal and ethical assessment
    financial and operational valuation
    uncertainty and sensitivity report
    reproducibility record
```

The procedure does not automatically resolve disagreements between welfare criteria. Such disagreements should be exposed in the report.

16.3 Data Quality and Missingness

Let $R_{jk} = 1$ indicate that outcome Y_{jk} is observed.

Missing data may be:

- missing completely at random;
- missing at random conditional on observed variables;
- missing not at random.

The distinction matters because institutional data may be selectively recorded. Individuals experiencing harm may be less likely to respond to surveys, and institutions may measure favourable outcomes more consistently than adverse ones.

No single imputation method eliminates the need to investigate the missingness mechanism.

16.4 Reproducible Research

A reproducible implementation should separate:

1. raw data;
2. cleaned data;
3. derived variables;
4. identification code;
5. statistical models;
6. classification rules;
7. sensitivity analysis;
8. output tables and figures.

Where data cannot legally or ethically be shared, the project should provide a clear description of data access restrictions and, where possible, a synthetic or de-identified demonstration dataset.

16.5 Model Validation

Predictive accuracy and causal validity are different properties.

A model may predict who exits an institution with high accuracy while failing to estimate the effect of the institution on welfare.

Validation should therefore include separate tests for:

- predictive performance;
- calibration;
- causal identification;
- measurement validity;
- subgroup performance;
- distributional accuracy;
- robustness under distribution shift.

No predictive metric alone establishes that an institutional classification is normatively justified.

17 Artificial Intelligence and Machine Learning

17.1 Behavioural Prediction

A machine-learning model may estimate the probability of an action:

$$\widehat{\pi}_{\psi}(a \mid s, I) = \Pr_{\psi}(A = a \mid S = s, I). \tag{128}$$

The model may be used to forecast participation, exit, compliance, learning, or other observable outcomes.

Its predictions should be evaluated on appropriate held-out data, with calibration and subgroup analysis where relevant.

A high-performing predictor does not establish that the predicted behaviour is beneficial, voluntary, or welfare-enhancing.

17.2 Reinforcement Learning

A reinforcement-learning formulation may represent an agent's policy as

$$\pi_\psi(a \mid s), \tag{129}$$

with a reward function $r(s, a)$ and a discounted objective

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t r(S_t, A_t) \right]. \tag{130}$$

This formulation is useful for describing a learning objective. However, an agent's reward function is not automatically a valid representation of human welfare.

If the reward function is misspecified, optimized behaviour may improve the numerical objective while worsening the intended outcome.

The model must therefore distinguish:

- agent reward;
- predicted human behaviour;
- causal institutional effects;
- human welfare;
- legally permissible actions.

17.3 AI-Assisted Institutional Evaluation

AI systems may assist with:

- document extraction;
- policy comparison;
- behavioural pattern detection;
- anomaly detection;
- forecasting;
- simulation;
- uncertainty summarization;
- sensitivity analysis;
- reproducibility checks.

High-stakes institutional decisions require meaningful human oversight, validation, documentation, and procedures for contesting errors.

The NIST AI Risk Management Framework provides a useful reference for organizing AI risk identification, measurement, management, and governance [7].

17.4 Model Risk and Distribution Shift

Let the training distribution be P_{train} and the deployment distribution be P_{deploy}.

A model trained under one institutional regime may fail when:

$$P_{\text{train}} \neq P_{\text{deploy}}. \tag{131}$$

Institutional rules, incentives, language, and populations can change substantially over time. Monitoring should therefore detect changes in input distributions, outcome relationships, missingness, and subgroup performance.

A system should be permitted to return an inconclusive result rather than forcing a classification when the data fall outside its validated domain.

18 Cybersecurity, Privacy, and Adversarial Behaviour

18.1 Institutional Data as a Security Asset

Institutional analysis may involve sensitive behavioural, financial, legal, medical, educational, or political information.

The system should therefore apply:

- data minimization;
- purpose limitation;
- access controls;
- encryption where appropriate;
- audit logging;
- retention limits;
- secure model deployment;
- incident response;
- data-subject rights where applicable.

Privacy is not merely an implementation detail. Surveillance may change behaviour and may itself affect welfare, autonomy, and trust.

18.2 Adversarial Data and Strategic Manipulation

Institutions and individuals may adapt strategically to measurement.

Examples include:

- optimizing performance indicators rather than underlying outcomes;
- selectively reporting favourable data;
- manipulating eligibility classifications;
- changing behaviour to evade monitoring;
- exploiting model weaknesses;
- suppressing complaints or exit records.

This is a general form of measurement-induced behavioural change.

A robust system should compare reported indicators against independent evidence, maintain audit trails, investigate anomalies, and avoid relying on a single metric where incentives to manipulate it are substantial.

18.3 Threat Modelling

Let $\mathcal{T}_i$ denote the threat set for institution I_i. A security assessment may examine the probability, severity, and recoverability of each threat.

A simple expected-loss representation is

$$EL_i = \sum_{k=1}^{K} p_{ik} L_{ik}, \tag{132}$$

where p_{ik} is the probability of threat k and L_{ik} is its loss under the specified model.

Expected loss does not fully capture catastrophic tail risks, uncertainty about probabilities, or the ethical significance of certain harms. It should therefore be complemented by stress tests, scenario analysis, and rights-based constraints.

19 An Interdisciplinary Integration Architecture

19.1 Fields and Their Analytical Roles

Table 2: Interdisciplinary Contributions to the General N-Institutions Model

Field	Primary contribution	Principal limitation
Institutional economics	Rules, governance, incentives, organizational forms	Institutional descriptions do not establish welfare effects
Behavioural economics	Choice, bounded rationality, framing, adaptation	Behaviour does not uniquely identify welfare
Causal inference	Counterfactual effects and identification	Results depend on assumptions and research design
Welfare economics	Evaluation of outcomes and distributions	Aggregation requires normative commitments
Econometrics	Estimation, uncertainty, heterogeneous effects	Statistical precision does not guarantee causal validity
Finance	Cash flows, risk, discounting, asset valuation	Financial value is not identical to social value
Law	Authority, rights, liability, due process	Legal validity does not guarantee effectiveness
Psychology and psychiatry	Mental states, stress, cognition, health	Measurement and diagnosis require appropriate expertise
Political science	Power, representation, legitimacy, collective choice	Formal institutions may differ from actual practice
Physics and complex systems	Dynamics, feedback, stability, network effects	Physical analogies do not establish social laws
Chemistry and biology	Stocks, flows, adaptation, ecological constraints	Biological analogies do not justify political arrangements
Computer science	Algorithms, data architecture, reproducibility	Automation does not resolve normative disagreement
AI and machine learning	Prediction, simulation, anomaly detection	Prediction does not establish causation or welfare
Cybersecurity	Threat modelling, privacy, system integrity	Risk scores cannot replace legal and ethical analysis
International relations	Cross-border effects, strategic interaction, diplomacy	National aggregates may conceal foreign externalities
Environmental science	Environmental exposures and long-term effects	Uncertainty and nonmarket valuation remain important

19.2 The Integration Rule

Each discipline contributes evidence or methods to a specified analytical question. No discipline should be used as a substitute for another discipline's central task.

For example:

- a psychological model may explain behaviour but not establish a causal treatment effect;
- an econometric model may estimate an effect but not decide the ethical weights assigned to different groups;
- a financial model may value cash flows but not determine the legitimacy of the underlying activity;
- a legal analysis may identify obligations but not establish that an institution achieves its intended outcomes;
- an AI model may predict behaviour but not define human welfare.

The integrated framework is strongest when each conclusion is supported by the appropriate method and its limitations are explicit.

20 A General Institutional Evaluation Protocol

20.1 Stage One: Define the Institution

Record:

1. the unit of analysis;
2. the institutional type and hybrid functions;
3. the legal form and jurisdiction;
4. the relevant period;
5. the declared objectives;
6. the decision structure;
7. the rules and constraints.

20.2 Stage Two: Map the Affected Population

Identify:

1. insiders and current participants;
2. outsiders and indirect beneficiaries;
3. excluded and rejected individuals;
4. former participants;
5. individuals unable to exit;
6. groups affected through spillovers;
7. future populations where relevant.

20.3 Stage Three: Measure Behaviour and Outcomes

Collect behavioural records and welfare-relevant indicators using methods appropriate to the institution and population.

Document:

- the source of each variable;
- the operational definition;
- missingness and measurement error;
- selection and coverage limitations;
- privacy and consent requirements;
- comparability across groups and periods.

20.4 Stage Four: Specify the Counterfactual

State precisely which alternative institutional arrangement is being compared.

The counterfactual should be substantively plausible and relevant to the decision under consideration.

Where several counterfactuals are plausible, estimate or discuss each rather than selecting only the one that produces the most favourable conclusion.

20.5 Stage Five: Identify Causal Effects

Choose an identification strategy that fits the data and intervention. State the assumptions, estimate uncertainty, and conduct appropriate diagnostic and sensitivity analyses.

If a credible causal effect cannot be identified, report descriptive associations separately from causal claims.

20.6 Stage Six: Classify Welfare Effects

Estimate the distribution of welfare effects separately for relevant populations.

Use prespecified thresholds and uncertainty-aware rules. Do not infer that all individuals benefit from a positive average.

20.7 Stage Seven: Evaluate Purpose, Law, and Legitimacy

Assess the institution's objectives, actual practices, legal duties, procedural safeguards, rights implications, and accountability mechanisms.

Keep these assessments distinct from welfare classification while examining their interactions.

20.8 Stage Eight: Perform Institution-Specific Valuation

Select methods appropriate to the institutional type:

- discounted cash flow for relevant financial claims;
- cost-effectiveness analysis where outcomes are measurable in common nonmonetary units;
- cost-benefit analysis where monetization is defensible;
- capability and distributional analysis for welfare;
- resilience and risk analysis for systemic institutions;
- rights-constrained assessment for legally restricted decisions;
- multi-criteria analysis where dimensions remain incommensurable.

20.9 Stage Nine: Report Uncertainty and Limitations

Every institutional report should disclose:

1. what was measured;
2. what was inferred;
3. which assumptions were necessary;
4. which populations were not adequately represented;
5. how results change under alternative specifications;
6. which conclusions remain uncertain;
7. what further evidence would change the classification.

21 Empirical Research Programme

21.1 Research Questions

The framework generates several testable questions.

RQ1: Behavioural information. How much do behavioural records improve prediction of institutionally relevant outcomes beyond conventional institutional indicators?

RQ2: Causal identification. Which research designs most reliably estimate welfare effects for different institutional types?

RQ3: Distributional heterogeneity. How frequently do institutional interventions benefit insiders while harming outsiders, or produce the reverse pattern?

RQ4: Counterfactual sensitivity. How often do classifications change when a different credible counterfactual is used?

RQ5: Measurement validity. Do welfare measures remain comparable across institutional types, cultures, languages, and populations?

RQ6: Long-run effects. How often do short-run institutional effects differ in sign or magnitude from long-run effects?

RQ7: Predictive versus causal performance. How well do machine-learning predictions perform when compared with causal estimates and independently measured welfare outcomes?

RQ8: Governance and rights. How frequently do financial or aggregate welfare improvements coexist with adverse rights or legitimacy assessments?

21.2 Proposed Validation Design

A validation programme should begin with a limited set of institution types for which data and defensible comparison designs are available.

For each application, the researchers should preregister:

- the estimand;
- the counterfactual;
- the target population;
- the welfare measures;
- the identification strategy;
- the classification thresholds;
- the primary outcomes;
- the uncertainty and sensitivity procedures.

The framework should then be tested across different institutional contexts rather than evaluated only on the examples used to design it.

21.3 Baseline Comparisons

The proposed model should be compared against:

1. financial-performance-only classifications;
2. stated-mission-only classifications;
3. observed-participation-only classifications;
4. unadjusted outcome comparisons;
5. conventional regression-based approaches;
6. the full counterfactual and distribution-sensitive model.

The comparison should examine classification stability, predictive performance, causal credibility, distributional coverage, and decision usefulness.

No improvement should be claimed until the empirical results support it.

21.4 Falsification and Failure Conditions

The proposed framework would require revision if empirical studies show that:

- its measurement architecture systematically misclassifies welfare effects;
- its inside–outside partition misses important affected populations;
- its classifications are dominated by arbitrary modelling choices;
- its causal estimates fail under credible validation designs;
- its data requirements make reliable use infeasible in important institutional contexts;
- its use systematically increases privacy, discrimination, or governance risks without compensating benefits.

A framework that cannot be revised in response to contrary evidence is not an adequate scientific framework.

22 Limitations and Open Problems

22.1 The Measurement Problem

Welfare is multidimensional and may not admit a universally accepted scalar measure.

The proposed framework can organize evidence without resolving all philosophical disputes about the nature of well-being.

22.2 The Identification Problem

Institutional exposure is often endogenous. Individuals select into institutions, institutions select participants, and both may respond to unobserved circumstances.

Some institutional effects may be impossible to identify precisely with available data.

22.3 The Boundary Problem

Inside–outside classification is useful but incomplete when membership is fluid, overlapping, or strategically defined.

A future extension should consider fuzzy membership, network exposure, and continuous dependence.

22.4 The Aggregation Problem

No universal aggregation rule can be presumed to satisfy every reasonable ethical requirement.

The choice of social weights, thresholds, and non-compensable constraints must remain visible.

22.5 The Reflexivity Problem

Institutions may change behaviour because they know they are being evaluated. Individuals may adapt to indicators, and administrators may optimize reported metrics.

The measurement process can therefore become part of the system being measured.

22.6 The Computational Problem

A system of N institutions may contain numerous interactions. If all pairwise interactions are considered, the number of possible pairs is

$$\binom{N}{2} = \frac{N(N-1)}{2}. \tag{133}$$

If all subsets of institutions are considered as possible interacting groups, there are

$$2^N - 1 \tag{134}$$

nonempty subsets.

These counts illustrate the rapid growth of the interaction space. They do not imply that every interaction is empirically meaningful or that every subset must be estimated.

Practical models will require theory-guided sparsity, modularity, hierarchical modelling, or other justified simplifications.

23 Conclusions

The General N-Institutions Model proposes a common procedure for analysing institutions whose organizational forms and purposes differ substantially.

Its foundational sequence is

$$\boxed{\text{Behaviour} \rightarrow \text{Causal Inference} \rightarrow \text{Welfare Effects} \rightarrow \text{Classification} \rightarrow \text{Institution-Specific Valuation.}} \tag{135}$$

Behaviour supplies evidence about how individuals respond to institutional arrangements. Causal inference estimates how outcomes differ under specified alternatives. Welfare analysis evaluates the resulting changes under explicit measures and ethical commitments. Classification summarizes the evidence without concealing distributional differences. Valuation then applies methods appropriate to the institution's objectives, constraints, legal status, and economic characteristics.

The model's central methodological contribution is to separate questions that are often conflated:

- what an institution is;
- what it claims to do;
- how people behave within and around it;
- what effects it causes;
- whose welfare increases or decreases;
- whether its conduct is lawful and legitimate;
- what its financial and social value may be.

The framework also distinguishes a shared analytical procedure from a shared substantive value scale. It permits the comparison of institutions without assuming that every institution has the same purpose or that every relevant outcome can be monetized.

Its mathematical propositions establish properties of the proposed formal structure, not universal empirical laws about institutional performance. The empirical claims remain hypotheses to be tested through transparent measurement, credible identification, robustness analysis, and replication.

The General N-Institutions Model should therefore be understood as a research architecture: a framework for asking better questions, making assumptions explicit, preserving distributional information, and identifying the evidence needed for defensible institutional judgements.

The governing principle is straightforward:

> Do not infer institutional value from institutional identity. Investigate behaviour, identify causal effects, evaluate welfare, preserve the differences among affected individuals, and only then undertake institution-specific valuation.

A Formal Definitions and Additional Results

A.1 Definition: Institutional Welfare Profile

For institution I_i, define the welfare profile

$$\boldsymbol{\Omega}_i(T) = \left(\mathcal{D}_i^{\text{in}}, \mathcal{D}_i^{\text{out}}, \mathcal{U}_i, \mathcal{R}_i\right), \tag{136}$$

where:

$$\mathcal{D}_i^{\text{in}} = \text{inside welfare-effect distribution}, \tag{137}$$

$$\mathcal{D}_i^{\text{out}} = \text{outside welfare-effect distribution}, \tag{138}$$

$$\mathcal{U}_i = \text{uncertainty characterization}, \tag{139}$$

$$\mathcal{R}_i = \text{relevant rights and constraints}. \tag{140}$$

This profile is the preferred descriptive object when a single scalar classification would discard important information.

A.2 Proposition 4: A Positive Average Does Not Establish a Positive Median

Proposition. A positive mean welfare effect does not imply a positive median welfare effect.

Proof. Consider five welfare effects:

$$(-2, -1, -1, -1, 10). \tag{141}$$

Their mean is

$$\overline{\Delta W} = \frac{-2 - 1 - 1 - 1 + 10}{5} = 1 > 0. \tag{142}$$

The median is -1, which is negative. Therefore a positive mean does not imply a positive median. □

A.3 Proposition 5: Classification Depends on the Counterfactual

Proposition. An institution's estimated welfare effect can change when the counterfactual changes, even if the observed outcome remains fixed.

Proof. Let observed welfare under the institution be $W(1) = 10$.

Under counterfactual A, suppose $W(0; A) = 8$. Then

$$\Delta W(A) = 10 - 8 = 2. \tag{143}$$

Under counterfactual B, suppose $W(0; B) = 12$. Then

$$\Delta W(B) = 10 - 12 = -2. \tag{144}$$

The observed treated outcome is identical, but the estimated effect changes sign. Thus the counterfactual is an essential component of the estimand. □

A.4 Proposition 6: Prediction Does Not Establish Causation

Proposition. Accurate prediction of an observed outcome does not, by itself, identify the causal effect of an institutional intervention.

Proof. Suppose a predictor estimates the conditional distribution $\Pr(W \mid D, X)$ accurately under the observed data-generating process.

Causal inference requires a quantity involving potential outcomes, such as

$$\mathbb{E}[W(1) - W(0)]. \tag{145}$$

The observational conditional distribution alone does not determine the joint distribution of the two potential outcomes or establish the assumptions needed to identify their difference.

Therefore predictive accuracy alone does not establish causal identification. □

A.5 Proposition 7: Financial Solvency Does Not Establish Positive Social Welfare

Proposition. A positive financial surplus does not logically imply a positive social welfare effect.

Proof. Let an institution's financial surplus be

$$\Pi = R - C > 0. \tag{146}$$

Let its social welfare effect be

$$\Delta SW = \Delta B - \Delta H, \tag{147}$$

where ΔB denotes welfare benefits and ΔH denotes welfare harms under a specified social evaluation rule.

The financial-surplus condition imposes no general mathematical restriction requiring $\Delta B > \Delta H$. It is therefore possible for $\Pi > 0$ while $\Delta SW < 0$. □

The proposition establishes logical non-equivalence, not the empirical claim that profitable institutions are generally harmful.

A.6 A Research Checklist

Before publishing an institutional classification, verify that:

1. the institutional unit and type are defined;
2. relevant inside and outside populations are identified;
3. excluded and former participants are considered;
4. the counterfactual is explicit;
5. welfare measures are justified;
6. behavioural indicators are not treated as welfare by definition;
7. the identification strategy is stated;
8. uncertainty and sensitivity are reported;
9. distributional effects are preserved;
10. legal and ethical constraints are examined;
11. financial value is separated from social welfare;
12. data privacy and security are addressed;
13. the results can be independently audited;
14. limitations and unresolved questions are disclosed.

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Glossary

Average Treatment Effect
The population-average difference between potential outcomes under two specified treatment regimes.

Behavioural Record
A time-indexed record of actions and relevant states used to study individual responses to institutional arrangements.

Behavioural Policy
A conditional distribution describing how an individual or agent selects actions given states, information, and constraints.

Capability
A substantive opportunity to achieve a valued functioning, rather than merely possession of a resource or a formal permission.

Causal Identification
The process of establishing conditions under which an estimand can be recovered from observed data.

Counterfactual
A specified alternative regime against which an observed or potential outcome is compared.

Distributional Effect
A change in the distribution of outcomes across individuals or groups, not merely a change in the mean.

Externality
An effect of an action or institution on individuals who are not fully represented in the decision-maker's private costs or benefits.

Institution
An organization or structured system of rules, authority, practices, and relationships that shapes human interaction.

Institutional Non-Equivalence Principle
The proposed methodological principle that a common analytical framework does not imply a common substantive valuation scale.

Institutional Profile
A structured representation of an institution's type, objectives, rules, authority, stakeholders, technologies, and environment.

Institutional Welfare Profile
The distribution of welfare effects inside and outside an institution, together with uncertainty and relevant constraints.

Potential Outcome
The outcome that would occur for a unit under a specified treatment or institutional regime.

Positivity
An identification condition requiring positive probabilities of relevant treatment regimes within the target covariate profiles.

Revealed Preference
An approach that infers preference information from observed choices, subject to assumptions about choice and the relation between choice and welfare.

Social Choice
The study of methods for converting individual preferences or evaluations into collective decisions or rankings.

Social Welfare Function
A rule that evaluates social states using individual outcomes and specified normative principles.

Stochastic Discount Factor
A random variable used in asset-pricing theory to represent the pricing of future payoffs.

Typology-First Approach
An approach that identifies an institution's organizational type, functions, constraints, and affected populations before selecting the relevant methods of evaluation.

Welfare Effect
A change in an explicitly defined welfare measure relative to a specified counterfactual.

Welfare-First Classification Principle
The proposed principle that institutional classification should be based on investigated welfare effects rather than inferred from institutional labels or declared purposes.

The End